# Logical Equivalences

Given propositions $p$, $q$, and $r$, a tautology $\mathbf{t}$, and a contradiction $\mathbf{c}$, the following logical equivalences hold.

1. *Commutative Laws:*

$$p \text{ AND } q \equiv q \text{ AND } p$$
$$p \text{ OR } q \equiv q \text{ OR } p$$

2. *Associative Laws:*

$$(p \text{ AND } q) \text{ AND } r \equiv p \text{ AND } (q \text{ AND } r)$$
$$(p \text{ OR } q) \text{ OR } r \equiv p \text{ OR } (q \text{ OR } r)$$

3. *Distributive Laws:*

$$p \text{ AND } (q \text{ OR } r) \equiv (p \text{ AND } q) \text{ OR } (p \text{ AND } r)$$
$$p \text{ OR } (q \text{ AND } r) \equiv (p \text{ OR } q) \text{ AND } (p \text{ OR } r)$$

4. *Identity Laws:*

$$p \text{ AND } \mathbf{t} \equiv p$$
$$p \text{ OR } \mathbf{c} \equiv p$$

5. *Negation Laws:*

$$p \text{ OR NOT } p \equiv \mathbf{t}$$
$$p \text{ AND NOT } p \equiv \mathbf{c}$$

6. *Double Negative Law:*

$$\text{NOT NOT } p \equiv p$$

7. *Idempotent Laws:*

$$p \text{ AND } p \equiv p$$
$$p \text{ OR } p \equiv p$$

8. *Universal Bound Laws:*

$$p \text{ OR } \mathbf{t} \equiv \mathbf{t}$$
$$p \text{ AND } \mathbf{c} \equiv \mathbf{c}$$

9. *De Morgan's Laws:*

$$\text{NOT } (p \text{ AND } q) \equiv \text{NOT } p \text{ OR NOT } q$$
$$\text{NOT } (p \text{ OR } q) \equiv \text{NOT } p \text{ AND NOT } q$$

10. *Absorption Laws:*

$$p \text{ OR } (p \text{ AND } q) \equiv p$$
$$p \text{ AND } (p \text{ OR } q) \equiv p$$

11. *Negation of* **t** *and* **c***:*

$$\text{NOT } \mathbf{t} \equiv \mathbf{c}$$
$$\text{NOT } \mathbf{c} \equiv \mathbf{t}$$

12. *Definition of Conditional:*

$$p \text{ IMPLIES } q \equiv \text{NOT } p \text{ OR } q$$

13. *Definition of Biconditional:*

$$p \text{ IFF } q \equiv (\text{NOT } p \text{ OR } q) \text{ AND } (\text{NOT } q \text{ OR } p)$$